

# A rule-3 disproof submission for conjecture 00000007121

## The Hirsch bound and its alleged “high-dimensional double configuration”

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### Abstract

Conjecture 00000007121 asserts, in a single sentence, that the Hirsch conjecture’s diameter bound *holds and* that a *counterexample* to that bound is the “high-dimensional double configuration”. Read literally, the statement conjoins a claim with its own negation, and is therefore contradictory. Independently, the Hirsch bound is known to be false: Santos (2012) constructed a 43-dimensional polytope with 86 facets and diameter  $44 > 86 - 43 = 43$ . We give both grounds, formalise the logical content, and state prominently the caveat that the entry may be a catalogue-style garbling rather than a substantive mathematical assertion.

## 1 The statement, quoted exactly

The catalogue entry `conjectures/00000007121.md` reads, verbatim:

**English.** Definition: Convex polytopes as geometric bodies of finite halfspace intersections. Conjecture: The Hirsch conjecture’s diameter upper bound of face count minus dimension holds, and the counterexample to the bound is the high-dimensional double configuration. (high-dimensional counterexample to the Hirsch bound)

**中文.** 定义：凸多面体：有限半空间交的几何体。猜想：Hirsch 猜想：图直径的上界为面数减维数且界的反例为高维双重构型。（Hirsch 反例高维双构型直径上界）

Let  $d$  denote the dimension,  $f$  the number of facets, and  $\text{diam}$  the graph diameter of a convex polytope. The Hirsch conjecture asserts

$$\text{diam} \leq f - d. \tag{1}$$

Write

$$A := (\forall d f \text{ diam}, \text{diam} \leq f - d), \quad B := (\exists d f \text{ diam}, f - d < \text{diam}).$$

The entry states  $A \wedge B$  (with the second conjunct further naming the “high-dimensional double configuration”).

## 2 Ground 1: the statement as written is self-contradictory

Over  $\mathbb{N}$  with truncating subtraction,  $f - d < \text{diam}$  is the negation of  $\text{diam} \leq f - d$  (Lean’s `Nat.not_1e`). Hence

$$B \iff \neg A,$$

and the statement as written is  $A \wedge \neg A$ . This is a statement-level (logical) refutation, and it is fully formalisable in core Lean 4; see `lean4/Main.lean`, where the headline result is

$$\text{hirsch\_as\_written\_false} : \neg(A \wedge B),$$

proved by a kernel-checkable `intro/exact` argument over `Nat`, together with the pure propositional fact  $\neg(P \wedge \neg P)$  and its instantiation.

**Proposition 1.** *The two conjuncts of conjecture 00000007121 cannot both hold:*

$$\neg(A \wedge B).$$

*Moreover any witness to  $B$  entails  $\neg A$ .*

*Proof.* For the first claim, suppose  $A \wedge B$ . From  $B$  obtain  $d, f, \text{diam}$  with  $f - d < \text{diam}$ . From  $A$  obtain  $\text{diam} \leq f - d$  for the same triple. Since natural subtraction yields  $\neg(\text{diam} \leq f - d)$  from  $f - d < \text{diam}$ , we reach a contradiction. The second claim is the same argument without the appeal to  $A$ . Both are formalised in `lean4/Main.lean` (`counterexample_refutes_bound`, `hirsch_as_written_false`, `conjecture_00000007121_false`).  $\square$

### 3 Ground 2: the Hirsch bound is genuinely false (literature)

Independently of the phrasing issue, (1) is false. Santos [1] constructed a convex polytope of dimension 43 with 86 facets whose graph diameter is 44. Since

$$86 - 43 = 43 < 44,$$

this violates (1) and disproves the Hirsch conjecture. The polynomial Hirsch conjecture remains open.

*Remark 2.* This is a published literature result. It cannot be reproduced locally, and this submission does not attempt to do so. The file `reproduce.py` checks only the arithmetic  $86 - 43 = 43 < 44$  and states explicitly that Santos’s construction itself is not reproduced. The Lean development does *not* formalise Santos’s polytope; it formalises only the logical inconsistency of the statement as written.

### 4 Caveat: the entry may be a catalogue artefact

We state this prominently and do not overstate the logical reading.

- The sentence names a conjecture and an alleged disproof of it in one breath. This is characteristic of an automatically assembled catalogue entry that has concatenated a statement with a “counterexample” tag, rather than a deliberate mathematical assertion.
- “High-dimensional double configuration” is **not** the standard name for Santos’s object. The standard terminology is *Hirsch polytopes*, *spindles*, and the *Klee–Walkup* and *Santos* constructions. The phrase appears to be a non-standard or garbled label.
- Consequently the “self-contradiction” of Ground 1 is arguably a *phrasing artefact* rather than a substantive mathematical falsehood, even though it is a genuine consequence of the text as written.

We therefore present both grounds—the literal logical inconsistency and the independently known mathematical falsity—and leave the maintainer to judge which reading of the entry is intended.

## References

- [1] F. Santos. A counterexample to the Hirsch conjecture. *Annals of Mathematics* **176** (2012), no. 1, 383–412.